Hyperspectral Anomaly Detection With Relaxed Collaborative Representation

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***Abstract*— Anomaly detection has become an important remote sensing application due to the abundant spectral and spatial information contained in hyperspectral images. Recently, hyperspectral anomaly detection methods based on the collaborative representation (CR) model have attracted significant attention. Nevertheless, these methods have to face two main challenges: 1) all features (spectral signatures) are constrained to share the same representation coefficient, which ignores the differences among features and 2) existing dictionaries for pixel-by-pixel detection models are usually not reliable. To address these issues, this article proposes a new relaxed CR detector for hyperspectral anomaly detection by using a novel nonglobal dictionary. The proposed detector conducts CR on each feature dimension of the pixel under test and simultaneously constrains the coding vectors of different features to be similar. To the best of our knowledge, this is the first time that a detection model is built from each feature dimension. To adjust the contributions of each feature, an adaptive feature weight constrained version of the method is also proposed. The nonglobal dictionary is constructed by combining the *k*-nearest neighbor method and an existing global dictionary, which is more reliable and practical than the widely used dual-window dictionary. In addition, this article also designs a band selection strategy for the proposed method. Experiments on five real datasets indicate that the proposed method suppresses background well and outperforms other classical and state-of-the-art methods.**

***Index Terms*— Collaborative representation (CR), dictionary, hyperspectral anomaly detection.**

I. Introduction

ANY hyperspectral image can be interpreted as a 3-D cube, which consists of two spatial dimensions and one spectral dimension. An important difference between hyperspectral images and other images is that the spectral response of a ground object provides a smooth curve (denoted as spectral signature) [1]. Such detailed spectral and spatial information provides the possibility of diagnostic recognition of ground objects [2]–[4]. Therefore, target detection has become a very important application of hyperspectral images, over a wide range of practical activities, such as military target detection, mineral identification, precision agriculture, change detection, and surveillance [5]–[9]. In this context, hyperspectral target detection is divided into *matched target detection* and *anomaly detection*, based on whether the target is known or not. In this regard, this work focuses on the latter.

A. Hyperspectral Anomaly Detection

Hyperspectral anomaly detection has attracted enormous attention due to its potential application, as it does not require any prior information about targets and backgrounds. Although the concept of anomaly has still not been defined exactly [10], it is generally accepted that an anomaly deviates from the background clutter and generally covers a small area significantly different from the surrounding background, occupying a small proportion of the image [11]. For instance, a plane in the sky, a man in disguise in the woods, or a rabbit in a field can be deemed as anomalous targets [12], [13]. Despite the fact that anomaly detection is an unsupervised dichotomous problem, the classification criterion based on the minimum overall classification error is generally not suitable in this context [6], [13]. In this sense, hyperspectral anomaly detection methods can be mainly divided into two categories: *statistics*- and *reconstruction-based* methods.

B. Statistics-Based Methods

Focusing on the former, statistics-based hyperspectral anomaly detection methods assume that the background distribution may obey some statistical rules, while anomalies may not. For instance, the Reed–Xiaoli (RX) method is a classical statistics-based approach [14] in which the mean and covariance of the original image are computed to estimate the probability density function of a Gaussian model, and then, pixels that deviate from the Gaussian distribution are marked as anomalies. This method generally exhibits good performance (without any parameters) for images with simple background

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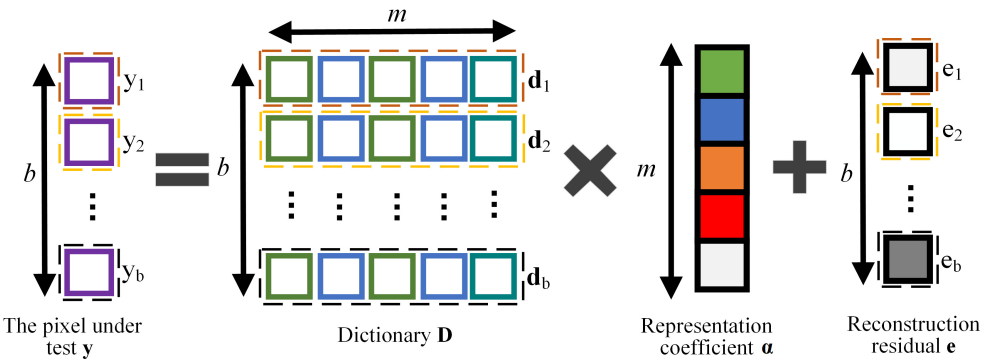


Fig. 1. Schematic of the CR model for hyperspectral anomaly detection.

distribution. Nonetheless, the Gaussian assumption is not appropriate for images with complex background distribution, and false alarms increase in the detection results [15]. Therefore, some advanced methods have been designed based on RX. For instance, the local RX (LRX) adopts a sliding dual window to obtain the surrounding background around the pixel under test [16], and the RX method is then used to determine whether the pixel under test belongs to the background or not. Compared with the original RX, i.e., global RX, LRX is more adaptive. On the other hand, the kernel RX (KRX) maps the original image into a higher dimensional feature space in which the separation between background and anomalies is larger than in the original feature space [17], and then, RX is implemented to detect anomalies. Another classical anomaly detection method is the support vector data description (SVDD), which utilizes a kernel function to map the original data into a higher feature space first [18] and then finds the smallest hypersphere supported by the background. The pixels out of the hypersphere are regarded as anomalies. Recently, new anomaly detectors have been proposed, such as the kernel isolation forest detector (KIFD) [19] and the detector based on the two-step generalized likelihood ratio test (2S-GLRT) [20]. These methods can also effectively detect anomalies by using some statistical knowledge.

C. Reconstruction-Based Methods

On the other hand, reconstruction-based hyperspectral anomaly detection methods assume that a normal pixel belonging to the background can be well reconstructed by a background dictionary, while an anomalous pixel cannot, and the reconstruction error is deemed as the anomaly degree of the pixel under test. Reconstruction-based methods include two important interrelated aspects: 1) the reconstruction model and 2) the dictionary. According to the different regularization constraints to the model, there are mainly three linear reconstruction models: the low-rank representation (LRR) model [21], [22], the sparse representation (SR) model [23], and the collaborative representation (CR) model [24]. Among them, LRR is a global constraint model, which regards the original image matrix as the sum of the background matrix and the anomaly matrix. The background matrix is constrained by low rank because most of the pixels in the image belong to the background, while the anomaly matrix is sparsely constrained as anomalous targets generally occupy a small portion of the image. Based on this idea, Xu *et al.* [25] proposed a hyperspectral anomaly detector based on LRR and SR (LRASR), which obtains good performance in small target detection. Cheng and Wang [26] proposed an anomaly detector that reduces the impact of noise on detection results by adding a regularized total variation constraint to LRR (GTVLRR). Wu *et al.* [27] proposed a scalable low-dimensional manifold model for anomaly detection (SLDMM-AD) that jointly exploits global and local information. In [28], a detector considering both global low-rank and local collaboration properties of the image is designed, which has been shown to be effective for both small and planar target detection. Similar to LRR, some anomaly detection methods based on tensor decomposition have been proposed in recent years. In this regard, tensor-based methods consider the hyperspectral cube as a third-order tensor, which can be interpreted as the composition of a background tensor and an anomaly tensor [29]–[32], where the anomaly tensor is the detection result map.

Sparse regression-based anomaly detection methods adopt several atoms in an overcomplete dictionary to reconstruct each pixel under test competitively by using the l_1 norm. The anomaly detector using background joint SR is a typical example [33], in which a triple sliding window strategy is used to build the background dictionary and finish representation. Different from SR, CR-based anomaly detection methods take advantage of the collaboration of atoms in a dictionary to represent each pixel under test by using the l_2 norm [24]. Since the first CR-based detector (CRD) was proposed [34], this kind of method has attracted lots of attention due to its simple computations and good performance. They allow every dictionary atom to participate in the representation of the pixel under test with a small coefficient, which is more suitable for anomaly detection tasks. However, the dictionary of the CRD was built by a dual sliding window. The contamination problem due to unsuitable dual-window sizes limits the performance of this method. Many strategies have been designed to solve this problem, such as eliminating the low probability points in the Gaussian distribution [35], spatial-domain principal component analysis (PCA) [36], fast density peak clustering [37]–[39], and fractional Fourier transform [40]. In addition to utilizing the spatial information,

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Imani [41] extracted morphological features to implement the CRD. Tan *et al.* [42] merged the detection results of multiple windows to improve the stability of the CRD. Zhao *et al.* [43] combined CRD with stacked autoencoders to mine deep feature for anomaly detection. In recent years, autoencoders have also been applied to anomaly detection (which can also be regarded as a type of reconstruction-based method) [44], [45]. This method detects anomalies by encoding and decoding the pixel under test, and the difference between the original input pixel and the encoded and decoded pixel is considered as an anomalous level of the pixel under test. In [46], a robust graph autoencoder (RGAE) for anomaly detection was proposed by embedding a supergraph in the autoencoder framework. This method outperforms traditional autoencoder detectors.

D. Statistics- Versus Reconstruction-Based Methods

By carefully analyzing these methods, we can conclude that statistics-based methods exhibit limited performance and application, due to the fact that statistical assumptions are generally unsuitable for complex scenarios. Furthermore, sparse regression and collaborative regression-based methods are more adaptive than LRR-based methods because they are computed in a pixel-by-pixel fashion, and collaborative regression-based methods perform better than sparse regression-based ones by including the collaboration of atoms in a dictionary. Nevertheless, if the CR model is analyzed from the viewpoint of feature dimensions, it is clear that each feature dimension of the pixel under test is constrained to share the same coding vector [47], [48], as depicted in Fig. 1. This can disregard some important features, which can better represent a certain class of pixels. Especially in complex application scenarios, the reasonable use of effective features can further improve the effectiveness and universality of the detectors. In addition, the background dictionaries of methods based on collaborative regression or sparse regression are built mostly by using multiple sliding windows, which are very sensitive to the size of anomalous targets. In this regard, without any prior information, the appropriate size of multiple windows is very difficult to be defined. As a result, this article focuses on the design of anomaly detectors based on relaxed CR (RCRD), which can flexibly use different features with an adaptive dictionary to achieve high-accuracy detection.

E. Proposed Contribution

In this article, inspired by Yang *et al.* [47], we develop a new hyperspectral anomaly detector based on RCRD and a new nonglobal dictionary. Different from the CRD, our approach conducts CR for the pixel under test on every spectral feature dimension, and it simultaneously constrains each representation coefficient close to the mean representation coefficient, which keeps the difference and similarity among features. In order to adjust the contributions of different features, we develop a constrained version of RCRD with a weight (RCRDW) by introducing an adaptive weight that is updated automatically in each iteration [47]. The new nonglobal dictionary is constructed by using the k-nearest neighbor (KNN) algorithm on an existing global dictionary [28], which adaptively selects k-nearest dictionary atoms for each pixel under test to participate in the representation. In addition, a band selection strategy for RCRD and RCRDW is also designed. The main innovative contributions of this article can be summarized as follows.

1) We develop a new anomaly detector based on RCRD, which can not only consider both the differences and similarities among spectral features but also achieve good results without iterative computation. Compared with CRD, RCRD has looser constraints on feature encoding and stronger representation capability. To the best of our knowledge, this is the first time that a detection model is built from each feature dimension.
2) A feature weight is introduced in RCRD that can adjust the contribution of different features adaptively in an iterative fashion. In this way, the RCRDW can use features more flexibly to represent pixels and obtain better detection results than RCRD. Different from [47], RCRDW is designed from a spectral feature perspective based on the characteristics of hyperspectral images rather than from a spatial perspective. In addition, a band selection strategy is designed for both RCRD and RCRDW to evaluate the influence of band selection on these methods.
3) A nonglobal dictionary is incorporated into RCRD and RCRDW, with the goal of building an adaptive dictionary for the optimal reconstruction of each pixel under test by using the KNN on an existing global dictionary. It can be regarded as a local version of the global dictionary [28], which is suitable for any pixel-by-pixel detection method, and more reliable and practical than the widely used dual-window dictionary. Therefore, the proposed dictionary construction method may provide ideas for subsequent local dictionary designs.

F. Article Organization

The rest of this article is organized as follows. Section II introduces the proposed RCRD, its weighted variant (RCRDW), the nonglobal dictionary, and the band selection strategy for both RCRD and RCRDW. Section III describes the experimental results obtained by the proposed methods and some state-of-the-art approaches on five real datasets. Section IV discusses and verifies the effectiveness of the newly proposed methods. Finally, Section V concludes this article with some remarks and hints at plausible future research lines.

II. Proposed Methodology

In order to address some limitations of the CRD method, i.e., strong dependence to feature coding and possibly unreliable dictionaries, a new *relaxed* CR detector is introduced in this article. It includes two anomaly detection models: RCRD and RCRDW, and a novel nonglobal dictionary. Furthermore, a band selection strategy is also designed for RCRD and RCRDW, which aims to exploit fewer features and minimize running time. The proposed methods are based on optimizations of the CRD method. On the one hand, they inherit the advantages of CR itself. On the other hand, our methods

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take into account the differences between features and make full use of these differences to seek the best reconstruction of the background. Moreover, the nonglobal dictionary is more reliable and practical than the widely used dual-window dictionary.

A. Relaxed Collaborative Representation Detector

Let a 3-D hyperspectral image be given by a 2-D matrix $\mathbf{Y} \in \mathbb{R}^{b\times N}$, where b is the number of bands and N is the number of pixels. In the CRD, each pixel $\mathbf{y}$ can be reconstructed by a linear combination of the background dictionary $\mathbf{D} \in \mathbb{R}^{b\times m}$ and the representation coefficient $\boldsymbol{\alpha} \in \mathbb{R}^{m\times 1}$, where m is the number of dictionary atoms, while the reconstruction error can be deemed as the anomaly degree of the pixel under test. The CRD is, thus, expressed as

$$\arg\min_{\hat{\boldsymbol{\alpha}}} \|\mathbf{y} - \mathbf{D}\boldsymbol{\alpha}\|_2^2 + \lambda\|\boldsymbol{\alpha}\|_2^2 \tag{1}$$

$$\mathbf{e} = \|\mathbf{y} - \hat{\mathbf{y}}\|_2 = \|\mathbf{y} - \mathbf{D}\hat{\boldsymbol{\alpha}}\|_2 \tag{2}$$

where λ is a regularization parameter, $\|\cdot\|_2$ is the l_2 norm, $\|\boldsymbol{\alpha}\|_2$ is a collaborative regularization constraint on $\boldsymbol{\alpha}$, $\mathbf{e}$ is the representation error of the pixel under test $\mathbf{y}$, $\hat{\mathbf{y}}$ is the prediction of $\mathbf{y}$, and $\hat{\boldsymbol{\alpha}}$ is the optimized value of $\boldsymbol{\alpha}$. Equation (1) is a convex function with an l_2 norm constraint and can be solved directly. When the CRD model is analyzed from the perspective of each spectral feature, as shown in Fig. 1, the pixel under test $\mathbf{y}$ is viewed as a stacking of the spectral response values of all bands, that is, $\mathbf{y} = [y_1; y_2; \ldots; y_b]$, and accordingly, the dictionary $\mathbf{D}$ is viewed as a stack of subdictionaries of all bands, that is, $\mathbf{D} = [\mathbf{d}_1; \mathbf{d}_2; \ldots; \mathbf{d}_b]$, where $\mathbf{d}_i \in \mathbb{R}^{1\times m}(i = 1, 2, \ldots, b)$ is the subdictionary of dictionary $\mathbf{D}$ on the ith feature, and the semicolon indicates the end of a row of the matrix. Then, (1) can be rewritten as

$$\arg\min_{\hat{\boldsymbol{\alpha}}} \sum_{i=1}^{b} \|y_i - \mathbf{d}_i\boldsymbol{\alpha}\|_2^2 + \lambda\|\boldsymbol{\alpha}\|_2^2. \tag{3}$$

The CRD constrains all spectral features with the same representation coefficient $\boldsymbol{\alpha}$. The goal of hyperspectral anomaly detection methods is to suppress background and highlight anomalies. Therefore, the reconstruction ability of the model is very important for background suppression. Hyperspectral images have hundreds of spectra, which finely draws the characteristics of different ground objects. The same ground object exhibits a different response in different bands, and the use of the same representation coefficient for all bands will inevitably weaken the representation ability of the model. Therefore, the differences between features should be taken into account when building a detection model to completely exploit the features. Specifically, every pixel under test should be expressed from each spectral feature perspective. That is to say, each feature y_i of the pixel under test $\mathbf{y}$ is linearly represented using the corresponding feature subdictionary $\mathbf{d}_i$, and the representation coefficient is $\boldsymbol{\alpha}_i$, where $i = 1, 2, \ldots, b$. The objective function is

$$\arg\min_{\hat{\boldsymbol{\alpha}}_i} \sum_{i=1}^{b} \|y_i - \mathbf{d}_i\boldsymbol{\alpha}_i\|_2^2 + \lambda\|\boldsymbol{\alpha}_i\|_2^2 \tag{4}$$

where $\hat{\boldsymbol{\alpha}}_i$ is the optimized value of $\boldsymbol{\alpha}_i$. In (4), each y_i is reconstructed by the corresponding representation coefficient $\boldsymbol{\alpha}_i$ and the subdictionary $\mathbf{d}_i$, while CRD constrains each y_i to be reconstructed by the subdictionary $\mathbf{d}_i$ and the same representation coefficient $\boldsymbol{\alpha}$, as shown in (1) and (3) and Fig. 1. Therefore, (4) provides a more relaxed constraint on the feature coding vector that is called the RCRD detector. To solve (4), every optimal $\boldsymbol{\alpha}_i$ is obtained by a closed solution, which may lead to significant differences of the values of $\boldsymbol{\alpha}_i$ for each spectral feature. The different features of a pixel are from the same entity, so the representation coefficients of different features from the same pixel should also have similarities to keep the same coding pattern. Therefore, a constraint is designed to ensure the coding similarity of different features as follows:

$$\arg\min_{\hat{\boldsymbol{\alpha}}_i} \sum_{i=1}^{b} \|\boldsymbol{\alpha}_i - \bar{\boldsymbol{\alpha}}\|_2^2 \tag{5}$$

where $\bar{\boldsymbol{\alpha}}$ is the average representation coefficient of all features. By constraining the minimum difference between $\boldsymbol{\alpha}_i(i = 1, 2, \ldots, b)$ and $\bar{\boldsymbol{\alpha}}$, the similarity of representation coefficients makes the representation stable. The objective function of RCRD taking into account the similarity and difference of features is

$$\arg\min_{\hat{\boldsymbol{\alpha}}_i} \sum_{i=1}^{b} \|y_i - \mathbf{d}_i\boldsymbol{\alpha}_i\|_2^2 + \lambda\|\boldsymbol{\alpha}_i\|_2^2 + \tau\|\boldsymbol{\alpha}_i - \bar{\boldsymbol{\alpha}}\|_2^2 \tag{6}$$

where λ and τ are tradeoff factors. Compared to the CRD, the RCRD model is designed from a feature perspective and simultaneously keeps the similarities of feature coding, thus taking the difference of features into account. Equation (6) is also solved directly by making the derivative equal to 0 as it was the case with the CRD. When the derivative of (6) equals 0, $\boldsymbol{\alpha}_i$ is expressed as

$$\hat{\boldsymbol{\alpha}}_i = \left(\mathbf{d}_i^{\mathrm{T}}\mathbf{d}_i + \mathbf{I}(\lambda + \tau)\right)^{-1}\left(\mathbf{d}_i^{\mathrm{T}}y_i + \tau\bar{\boldsymbol{\alpha}}\right) \tag{7}$$

where $(\mathbf{d}_i^{\mathrm{T}}\mathbf{d}_i + \mathbf{I}(\lambda + \tau))^{-1}$ is the inverse of $\mathbf{d}_i^{\mathrm{T}}\mathbf{d}_i + \mathbf{I}(\lambda + \tau)$. For the sake of simplicity, auxiliary variables $\boldsymbol{\alpha}_{0,i}$ and $\mathbf{P}_i$ are used, where

$$\boldsymbol{\alpha}_{0,i} = \left(\mathbf{d}_i^{\mathrm{T}}\mathbf{d}_i + \mathbf{I}(\lambda + \tau)\right)^{-1}\mathbf{d}_i^{\mathrm{T}}y_i \tag{8}$$

$$\mathbf{P}_i = \left(\mathbf{d}_i^{\mathrm{T}}\mathbf{d}_i + \mathbf{I}(\lambda + \tau)\right)^{-1} \tag{9}$$

where $\boldsymbol{\alpha}_{0,i}$ can be regarded as the initialization of $\boldsymbol{\alpha}_i$. Then, (7) can be rewritten as

$$\begin{aligned}\hat{\boldsymbol{\alpha}}_i &= \boldsymbol{\alpha}_{0,i} + \tau\mathbf{P}_i\bar{\boldsymbol{\alpha}} \\ &= \boldsymbol{\alpha}_{0,i} + \tau\mathbf{P}_i\frac{\sum_{i=1}^{b}\hat{\boldsymbol{\alpha}}_i}{b}.\end{aligned} \tag{10}$$

Furthermore, $\sum_{i=1}^{b}\hat{\boldsymbol{\alpha}}_i$ can be expressed as

$$\begin{aligned}\sum_{i=1}^{b}\hat{\boldsymbol{\alpha}}_i &= \sum_{i=1}^{b}\left(\boldsymbol{\alpha}_{0,i} + \tau\mathbf{P}_i\frac{\sum_{i=1}^{b}\hat{\boldsymbol{\alpha}}_i}{b}\right) \\ &= \sum_{i=1}^{b}\boldsymbol{\alpha}_{0,i} + \frac{\sum_{i=1}^{b}\tau\mathbf{P}_i}{b}\sum_{i=1}^{b}\hat{\boldsymbol{\alpha}}_i.\end{aligned} \tag{11}$$

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Then, $\sum_{i=1}^{b} \boldsymbol{\alpha}_i$ can be derived by (11) as

$$\sum_{i=1}^{b} \hat{\boldsymbol{\alpha}}_i = \left(\mathbf{I} - \frac{\sum_{i=1}^{b} \tau \mathbf{P}_i}{b} \right)^{-1} \sum_{i=1}^{b} \boldsymbol{\alpha}_{0,i}. \tag{12}$$

Therefore, (10) is rewritten by using (12) as

$$\hat{\boldsymbol{\alpha}}_i = \boldsymbol{\alpha}_{0,i} + \frac{\tau \mathbf{P}_i}{b} \left(\boldsymbol{I} - \frac{\sum_{i=1}^{b} \tau \mathbf{P}_i}{b} \right)^{-1} \sum_{i=1}^{b} \boldsymbol{\alpha}_{0,i}. \tag{13}$$

When all $\boldsymbol{\alpha}_i (i = 1, 2, \ldots, b)$ are obtained, the reconstruction error of the pixel under test $\mathbf{y}$ can be calculated by

$$\mathbf{e} = \sum_{i=1}^{b} \| y_i - \mathbf{d}_i \hat{\boldsymbol{\alpha}}_i \|_2^2. \tag{14}$$

The anomaly map $\mathbf{E} \in \mathbb{R}^{1 \times N}$ can be obtained through Algorithm 1.

Algorithm 1 Pseudocode of RCRD Model

Input:

Hyperspectral dataset: $\mathbf{Y}$
Dictionary: $\mathbf{D}$
Parameters: λ and τ

1: **for** each pixel under test $\mathbf{y}$ **do**
2: Compute $\boldsymbol{\alpha}_{0,i}$ and $\mathbf{P}_i$ by Eqs. (8-9)
3: Compute $\boldsymbol{\alpha}_i$ by Eq. (13)
4: Compute the reconstruction error of $\mathbf{y}$ by Eq. (14)
5: **end for**

Output:

Anomaly map matrix: $\mathbf{E}$

B. Relaxed Collaborative Representation Detector With an Adaptive Weight

Although the above RCRD can use the difference and similarity of features simultaneously to seek the best representation of the pixel under test, it is not difficult to infer that the RCRD model assigns the same weight to different features. In the RCRD model, the default weight of different features is 1, i.e., both noisy bands and bands that are rich in terms of information make equal contributions to the representation, which is not reasonable. In addition, different ground object types respond differently to different bands. Typical examples are vegetation and water indices, where the former is usually computed by using red and near-infrared bands, while the latter is usually computed by using green and infrared bands. Therefore, to make better use of valuable features for specific ground objects, an adaptive feature weight is added to the RCRD model (RCRDW) according to [47]. The RCRDW is given by

$$\begin{aligned} \arg\min_{\hat{\boldsymbol{\alpha}}_i, \omega_i} & \sum_{i=1}^{b} \| y_i - \mathbf{d}_i \boldsymbol{\alpha}_i \|_2^2 + \lambda \| \boldsymbol{\alpha}_i \|_2^2 \\ & + \tau \omega_i \| \boldsymbol{\alpha}_i - \bar{\boldsymbol{\alpha}} \|_2^2 \quad \text{s.t.} \quad prior\{\omega_i\} \end{aligned} \tag{15}$$

where ω_i is the weight of the ith feature, ensuring the different contribution of features to the representation. For hyperspectral anomaly detection purposes, the prior information of $\omega_i (i = 1, 2, \ldots, b)$ is hard to obtain. Thus, ω_i with a weak prior is computed by regularizing the entropy according to [47], that is,

$$-\sum_{i=1}^{b} \omega_i \ln(\omega_i) > \sigma \tag{16}$$

where σ is a threshold. Then, the RCRDW model is written according to (15) and (16) as follows:

$$\begin{aligned} \arg\min_{\hat{\boldsymbol{\alpha}}_i, \omega_i} & \sum_{i=1}^{b} \| y_i - \mathbf{d}_i \boldsymbol{\alpha}_i \|_2^2 + \lambda \| \boldsymbol{\alpha}_i \|_2^2 \\ & + \tau \omega_i \| \boldsymbol{\alpha}_i - \bar{\boldsymbol{\alpha}} \|_2^2 + \gamma \omega_i \ln(\omega_i) \end{aligned} \tag{17}$$

where γ is a tradeoff factor. It can be seen that the unknown variables in (17) include the coefficient matrix $\boldsymbol{\alpha}_i$, the weight ω_i, and the mean coefficient matrix $\bar{\boldsymbol{\alpha}}$, so it can be solved by iteratively updating $\boldsymbol{\alpha}_i$, $\bar{\boldsymbol{\alpha}}$, and ω_i in turn. Through continuous iteration, $\boldsymbol{\omega} = [\omega_1, \omega_2, \ldots, \omega_b]$ adjusts the contribution of different features in a flexible manner. The detailed solution is given as follows.

1) Fixing ω_i and Updating $\boldsymbol{\alpha}_i$: According to (7), $\boldsymbol{\alpha}_i$ in RCRDW is expressed as

$$\hat{\boldsymbol{\alpha}}_i = \left(\mathbf{d}_i^{\mathrm{T}} \mathbf{d}_i + \mathbf{I}(\lambda + \tau \omega_i) \right)^{-1} \left(\mathbf{d}_i^{\mathrm{T}} y_i + \tau \omega_i \bar{\boldsymbol{\alpha}} \right) \tag{18}$$

and then auxiliary variables $\boldsymbol{\alpha}_{0,i}$ and $\mathbf{P}_i$ are

$$\boldsymbol{\alpha}_{0,i} = \left(\mathbf{d}_i^{\mathrm{T}} \mathbf{d}_i + \mathbf{I}(\lambda + \tau \omega_i) \right)^{-1} \mathbf{d}_i^{\mathrm{T}} y_i \tag{19}$$

$$\mathbf{P}_i = \left(\mathbf{d}_i^{\mathrm{T}} \mathbf{d}_i + \mathbf{I}(\lambda + \tau \omega_i) \right)^{-1}. \tag{20}$$

Then, $\hat{\boldsymbol{\alpha}}_i$ is expressed as

$$\begin{aligned} \hat{\boldsymbol{\alpha}}_i &= \boldsymbol{\alpha}_{0,i} + \tau \omega_i \mathbf{P}_i \bar{\boldsymbol{\alpha}} \\ &= \boldsymbol{\alpha}_{0,i} + \tau \omega_i \mathbf{P}_i \frac{\sum_{i=1}^{b} \omega_i \hat{\boldsymbol{\alpha}}_i}{\sum_{i=1}^{b} \omega_i} \end{aligned} \tag{21}$$

where $\sum_{i=1}^{b} \omega_i \hat{\boldsymbol{\alpha}}_i$ is derived according to (11) and (12) as

$$\sum_{i=1}^{b} \omega_i \hat{\boldsymbol{\alpha}}_i = \left(\mathbf{I} - \sum_{i=1}^{b} \bar{\omega}_i \mathbf{P}_i \right)^{-1} \sum_{i=1}^{b} \omega_i \boldsymbol{\alpha}_{0,i} \tag{22}$$

where $\bar{\omega}_i = (\omega_i \tau \omega_i / \sum_{i=1}^{b} \omega_i)$. Then, (21) is rewritten by using (22) as

$$\hat{\boldsymbol{\alpha}}_i = \boldsymbol{\alpha}_{0,i} + \frac{\tau \omega_i \mathbf{P}_i}{\sum_{i=1}^{b} \omega_i} \left(\mathbf{I} - \sum_{i=1}^{b} \bar{\omega}_i \mathbf{P}_i \right)^{-1} \sum_{i=1}^{b} \omega_i \boldsymbol{\alpha}_{0,i}. \tag{23}$$

2) Updating $\bar{\boldsymbol{\alpha}}$:

$$\bar{\boldsymbol{\alpha}} = \frac{\sum_{i=1}^{b} \omega_i \hat{\boldsymbol{\alpha}}_i}{\sum_{i=1}^{b} \omega_i}. \tag{24}$$

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3) Fixing $\boldsymbol{\alpha}_i$ and Updating ω_i: The objective function of (17) is rewritten as

$$\arg\min_{\omega_i} \sum_{i=1}^{b} \tau\omega_i \|\boldsymbol{\alpha}_i - \bar{\boldsymbol{\alpha}}\|_2^2 + \gamma\omega_i \ln(\omega_i). \tag{25}$$

Let the derivative of (25) equal 0; then, ω_i is obtained by

$$\omega_i = \exp\left(-1 - \tau\|\hat{\boldsymbol{\alpha}}_i - \bar{\boldsymbol{\alpha}}\|_2^2/\gamma\right). \tag{26}$$

Equations (23), (24), and (26) are repeated until the solution of RCRDW converges, and the convergence condition is

$$\|\boldsymbol{\omega}^{k+1} - \boldsymbol{\omega}^k\|_2/\|\boldsymbol{\omega}^k\|_2 < \eta \tag{27}$$

where $\boldsymbol{\omega}^k$ is $\boldsymbol{\omega}$ in the kth iteration and η is a threshold, which is set as 1e-3, and generally two or three iterations will get to the threshold. When the stopping criterion is satisfied, $\boldsymbol{\alpha}_i$ and ω_i can be obtained. Then, the reconstruction error of each pixel under test $\mathbf{y}$ is calculated by (28) as

$$\mathbf{e} = \sum_{i=1}^{b} \omega_i \|y_i - \mathbf{d}_i\hat{\boldsymbol{\alpha}}_i\|_2^2. \tag{28}$$

The detailed procedure of RCRDW is summarized in Algorithm 2

Algorithm 2 Pseudocode of RCRDW Model

Input:

Hyperspectral dataset: $\mathbf{Y}$
Dictionary: $\mathbf{D}$
Parameters: λ, τ, and γ
Initialization of weight $\boldsymbol{\omega} = \mathbf{1}$

for each pixel under test $\mathbf{y}$ **do**
  Compute $\boldsymbol{\alpha}_{0,i}$ and $\mathbf{P}_i$ by Eqs. (19-20)
  while $\|\boldsymbol{\omega}^{k+1} - \boldsymbol{\omega}^k\|_2/\|\boldsymbol{\omega}^k\|_2 > \eta$ **do**
    Update $\hat{\boldsymbol{\alpha}}_i$ by Eq. (23)
    Update $\bar{\boldsymbol{\alpha}}$ by Eq. (24)
    Update ω_i by Eq. (26)
    Check the convergence condition in Eq. (27)
  end while
  Compute the reconstruction error of $\mathbf{y}$ by Eq. (28)
end for

Output:

Anomaly map matrix: $\mathbf{E}$

C. Nonglobal Dictionary Construction

CRD has become a benchmark algorithm due to its high detection rate. By using an appropriate dual window (with the inner window size being similar to the size of the targets and the outer window size being as small as possible) to build its local dictionary, CRD can generally reach a better performance than other methods [11]. However, in scenarios with large targets and relatively complex backgrounds, the performance of CRD is often limited due to the fact that an ideal size of the dual windows is difficult to set. Suboptimal dictionaries not only introduce outliers to the background dictionary but also reduce the optimal reconstruction of background pixels.

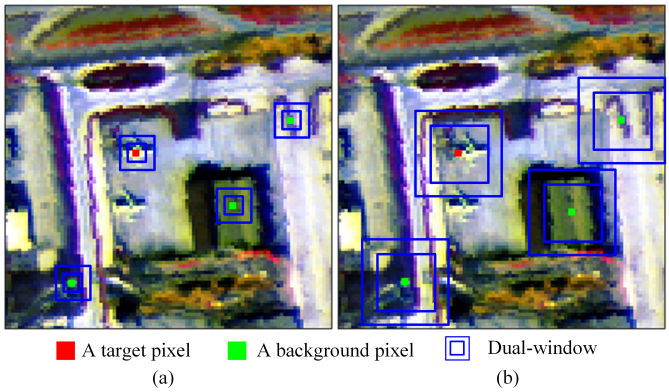


Fig. 2. Construction of the dictionary of a large target scene by using a dual window. (a) Small-size dual window. (b) Large-size dual window.

As shown in Fig. 2, a small-size dual window used for detecting large targets can introduce outliers' contamination when the pixel under test is anomalous. Although increasing the size of the dual windows can avoid outliers' contamination when detecting large targets, it cannot reconstruct normal pixels well. To address this problem, some outlier removal strategies have been developed, which mitigates the problem to some extent but requires tuning parameters.

In this section, a nonglobal dictionary construction method is designed especially for pixel-by-pixel anomaly detection models. It is also the localized version of a global dictionary construction method designed for a global detection model in [28]. The global dictionary is built by using twice mean shift clustering (2MSC), which not only includes almost all background classes but also reduces the contamination of outliers. In fact, this method is more practical as it does not require any parameter tuning. If the 2MSC global dictionary is directly used with our RCRD to detect anomalies, the following shortcomings are identified. On the one hand, too many atoms in the dictionary are found, leading to unstable solutions. On the other hand, the advantages of a pixel-by-pixel anomaly detection model will not be exploited. To address these issues, we apply the KNN algorithm to the 2MSC global dictionary to design a nonglobal dictionary that supports the proposed models: RCRD and RCRDW. It selects k-nearest dictionary atoms to build an adaptive dictionary for pixel under test by computing the Euclidean distance between the pixel $\mathbf{y}$ and the atoms in the 2MSC global dictionary $\mathbf{D}_{2\text{MSC}} \in \mathbb{R}^{b\times m_{2\text{MSC}}}$, where $m_{2\text{MSC}}$ is the number of atoms in $\mathbf{D}_{2\text{MSC}}$

$$\mathbf{dis} = \|\mathbf{D}_{2\text{MSC}}^i - \mathbf{y}\|_2^2, \quad i = 1, 2, \ldots, m_{2\text{MSC}}. \tag{29}$$

By calculating (29) and sorting $\mathbf{dis}$ in ascending order, the k dictionary atoms that most resemble the pixel under test are selected to form an adaptive dictionary $\mathbf{D}_{\text{NG}} \in \mathbb{R}^{b\times k}$. In this way, a few dictionary atoms not only make the solution stable but also express each pixel well with similar dictionary atoms. A flowchart of RCRD and RCRDW with a nonglobal dictionary is illustrated in Fig. 3.

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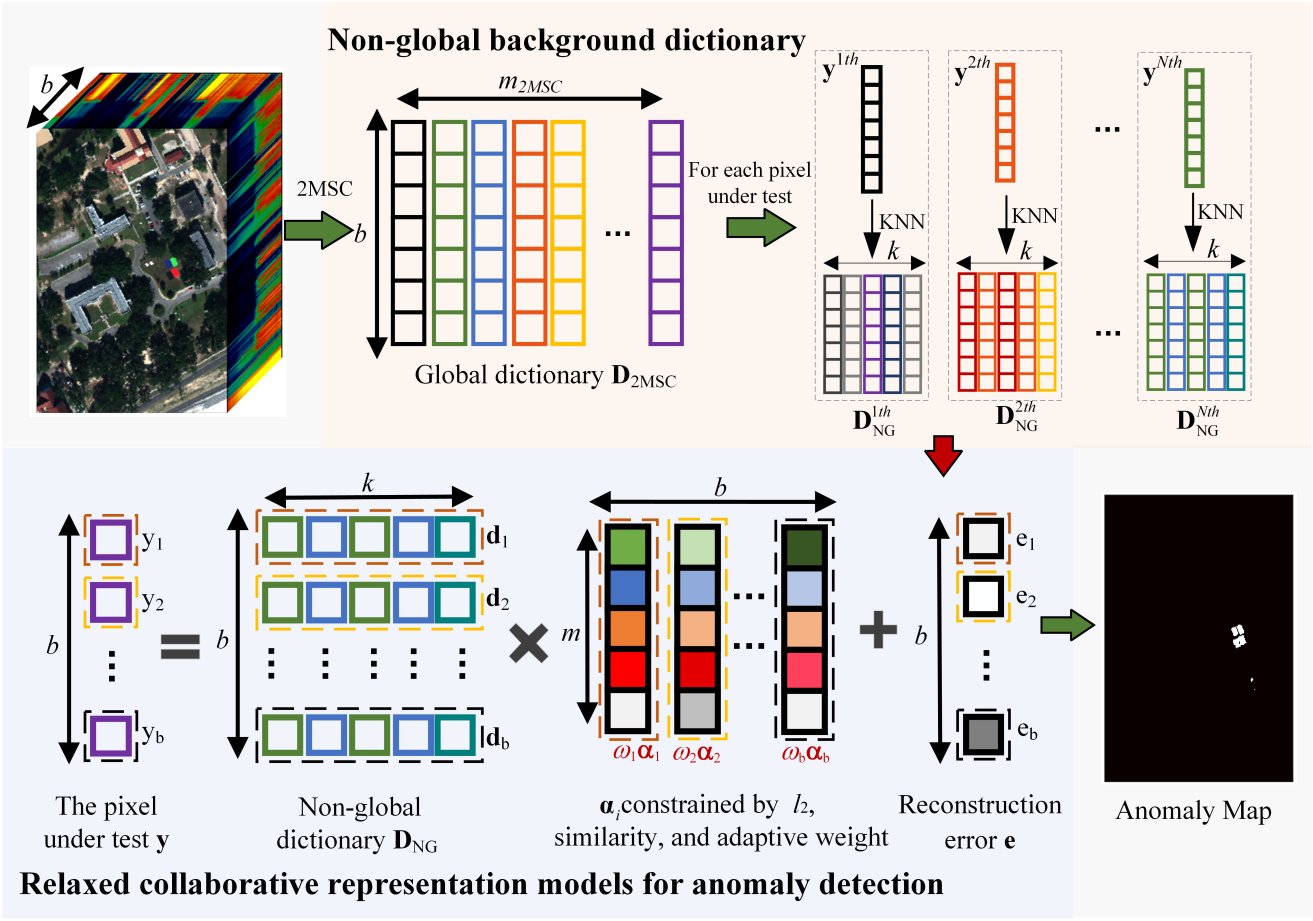


Fig. 3. Schematic of RCRD and RCRDW with a nonglobal dictionary.

D. Band Selection for RCRD and RCRDW

The proposed detection models (RCRD and RCRDW) obtain the representation coefficients of each feature dimension one by one to ensure the variability of different feature representations. At the same time, they constrain the difference between the representation coefficients and the average representation coefficients to be small to ensure the similarity of different feature codes. However, the bands of hyperspectral images generally comprise tens or even hundreds of wavelengths, and RCRD and RCRDW computed in pixel-by-pixel and feature-by-feature fashion would lead to a significant increase in computational effort. In addition, high redundancy between features may be detrimental to achieving an optimal linear representation. Therefore, this section further designs a band selection strategy for RCRD and RCRDW, pursuing high detection accuracy with less running time.

Hyperspectral image dimensionality reduction research includes two aspects: feature selection and feature extraction. The former preserves the original feature information, such as the minimum estimated abundance covariance (MEAC) method [49]. The latter rearranges and combines all features, such as PCA [50]. In addition, experimental results indicate that unsupervised methods are likely to discard information important to the anomalous targets due to the small percentage of anomalous targets in the image. In this sense, to better preserve the original information of the image, we use the MEAC method for dimensionality reduction, which is characterized by high accuracy, fast computation, and flexibility, and is widely used for feature selection purposes [51].

For simplicity, the mean shift clustering (MSC) method is adopted to obtain the spectral features of typical ground objects, which is also one part of the nonglobal dictionary construction. The spectra of cluster centers are used as the representative features of each class. Then, MEAC is used to find partial bands, which can retain the image information as much as possible. Let us assume that the spectral features matrix of typical ground objects in the image is $\mathbf{S} = [\mathbf{s}_1, \mathbf{s}_1, \ldots, \mathbf{s}_p]$, where p is the number of spectral features of typical ground objects. Every pixel can be expressed as $\mathbf{y} = \mathbf{S}\boldsymbol{\alpha}+\mathbf{n}$, where $\mathbf{n}$ is a noise term. By using the least-squares method, the estimate of $\hat{\boldsymbol{\alpha}}$ is written as $\hat{\boldsymbol{\alpha}} = (\mathbf{S}^{\mathrm{T}}\mathbf{S})^{-1}\mathbf{S}^{\mathrm{T}}\mathbf{y}$. Intuitively, the selected bands should make the difference between $\boldsymbol{\alpha}$ and $\hat{\boldsymbol{\alpha}}$ as small as possible, that is, the trace of $\mathrm{Cov}(\hat{\boldsymbol{\alpha}})$ should be as small as possible

$$\arg\min_{\boldsymbol{\Phi}} \left\{ \mathrm{trace}\left[\left(\hat{\mathbf{S}}^{\mathrm{T}}\hat{\mathbf{S}} \right)^{-1} \right] \right\} \tag{30}$$

where $\mathrm{Cov}(\hat{\boldsymbol{\alpha}})$ is the covariance of $\hat{\boldsymbol{\alpha}}$, $\mathrm{trace}[(\hat{\mathbf{S}}^{\mathrm{T}}\hat{\mathbf{S}})^{-1}]$ is the trace of $(\hat{\mathbf{S}}^{\mathrm{T}}\hat{\mathbf{S}})^{-1}$, $\boldsymbol{\Phi}$ is a subset of selected bands, and $\hat{\mathbf{S}}$ is the matrix of $\mathbf{S}$ on $\boldsymbol{\Phi}$. MEAC outputs an optimal bands subset by

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Algorithm 3 Pseudocode of RCRD and RCRDW With Band Selection and a Nonglobal Dictionary

Input:

Hyperspectral dataset: $\mathbf{Y} \in \mathbb{R}^{b\times N}$
Number of k-nearest neighbors: k
Number of selected bands: q
Parameters: λ, τ, and γ

$\mathbf{D}_{2\text{MSC}} \in \mathbb{R}^{b\times m_{2\text{MSC}}}$ and $\mathbf{S}$ are obtained by using the 2MSC global dictionary construction method in [28]
$\mathbf{\Phi}$ is found by Eq. (30)
Update $\mathbf{Y} \in \mathbb{R}^{b\times N}$ as $\mathbf{Y} \in \mathbb{R}^{\mathbf{\Phi}\times N}$, and $\mathbf{D}_{2\text{MSC}}$ as $\mathbf{D}_{2\text{MSC}} \in \mathbb{R}^{\mathbf{\Phi}\times m_{2\text{MSC}}}$
for each pixel under test **y do**
  Construct $\mathbf{D}_{\text{NG}}$ by Eq. (29)
  Execute Algorithm 1 or Algorithm 2 to detect anomalies
end for

Output:

Anomaly map matrix: **E**

traversing bands with (30). The image data and the dictionary on $\mathbf{\Phi}$ are input to RCRD or RCRDW to finish anomaly detection. The pseudocode of RCRD and RCRDW with band selection and nonglobal dictionary is given in Algorithm 3.

III. Experimental Results

In this section, five real hyperspectral image datasets with different characteristics (size of targets, image size, and complexity of background) and eight comparison methods are used to test the effectiveness of the newly proposed methods (RCRD and RCRDW). For evaluating the detection results, receiver operating characteristic (ROC) curves and box plots are adopted, where the former represents the accuracy of detection rate under different false alarm rates and the overall accuracy of the algorithm, and the latter displays the separation degree between the background and the anomalies, and the suppression degree of the background.

A. Experimental Datasets

The following datasets have been considered in experiments.

1) The first dataset is a small scene with a complex background for large target detection, which was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) on November 9, 2011 [52]. The image size is 100 × 100 pixels with 205 effective bands and 7.1-m spatial resolution. It covers a scene of Los Angeles, including two airplanes that are regarded as the anomalies of this dataset, as shown in Fig. 4.
2) The second dataset is a large scene with a relatively simple background for large target detection. It is collected by Nuance Cri sensor with 10-nm spectral resolution [53]. Different from the first dataset, this one includes 400 × 400 pixels and 46 effective bands. As shown in Fig. 4, ten rocks on the grassland are regarded as the anomalies of this dataset.
3) The third dataset is a large scene with a relatively complex background for large target detection, which was taken in November 2010 covering the University of Southern Mississippi Gulf Park campus (MUUFL) [54]. This dataset comprises 325 × 220 pixels with 64 effective bands.[1] As shown in Fig. 4, several cloth panels composed of 279 pixels are regarded as the anomalies of this dataset.
4) The fourth dataset is a small scene with a relatively complex background for small target detection, which was taken by the Hyperspectral Digital Imagery Collection Experiment HYDICE) in August 1995 covering an urban area [55]. The image includes 80 × 100 pixels and 175 effective bands. The spatial resolution is 1.56 m, and the spectral resolution is 10 nm. As shown in Fig. 4, several roofs and cars composed of 21 pixels in the image are regarded as anomalies.
5) The fifth dataset is a small scene with a relatively simple background for small target detection, which was collected by the Reflective Optics System Imaging Spectrometer ROSIS) over Pavia Center, Italy. The image size is 105 × 120 pixels with 102 effective bands and 1.3-m spatial resolution. As shown in Fig. 4, the background includes bridge, river, and bare soil, and the vehicles on the bridge are the anomalies of this dataset.

B. Algorithms Used for Comparison and Experimental Settings

Eight well-established methods are used to verify the performance of the proposed RCRD and RCRDW, including two classical methods, i.e., RX [14] and CRD [34], and six state-of-the-art methods, i.e., GTVLRR [26], SLDMM-AD [27], KIFD [19], 2S-GLRT [20], PTA [29], and RGAE [46]. Among them, RX, KIFD, and 2S-GLRT are based on the statistical theory; CRD, GTVLRR, SLDMM-AD, PTA, and RGAE are reconstruction-based models; and RGAE is based on deep learning architectures.

The parameters of proposed methods are λ, τ, γ, and k for nonglobal dictionary construction, where γ is the parameter of RCRDW. The setting range of λ and τ is {1e-5, 1e-4, 1e-3, 1e-2, 1e-1, 1e0, 1e1, 1e2}, and γ is {1e-4, 1e-3, 1e-2, 1e-1, 1e0, 1e1, 1e2, 1e3, 1e4}. In this section, k is set to 10 for every dataset except CRi, which only has one dictionary atom in the global dictionary. In the compared methods, the setting range of the dual-window size of CRD and 2S-GLRT is {(15,13), (15,11), (15,9), (15,7), (15,5), (15,3)}. The setting range of the regularization parameter of CRD is {1e-6, 1e-5, 1e-4, 1e-3, 1e-2, 1e-1, 1e0}. The other parameters of GTVLRR, SLDMM-AD, and KIFD are set to the values suggested in the corresponding papers. The parameters of PTA are μ, *rank*, α, β, and τ. All hyperparameters are set in {1e-4, 1e-3, 1e-2, 1e-1, 1e0, 1e1, 1e2, 1e3, 1e4}, and *rank* is set below 20. The setting range of the regularization parameter of RGAE is {1e-4, 1e-3, 1e-2, 1e-1}. The optimal result of each method is taken as the final detection result for the five considered experimental datasets. The detailed parameters' settings are listed in Table I, where the optimal values of λ, τ, and k of RCRDW are the same to RCRD.

[1] https://github.com/GatorSense/MUUFLGulfport

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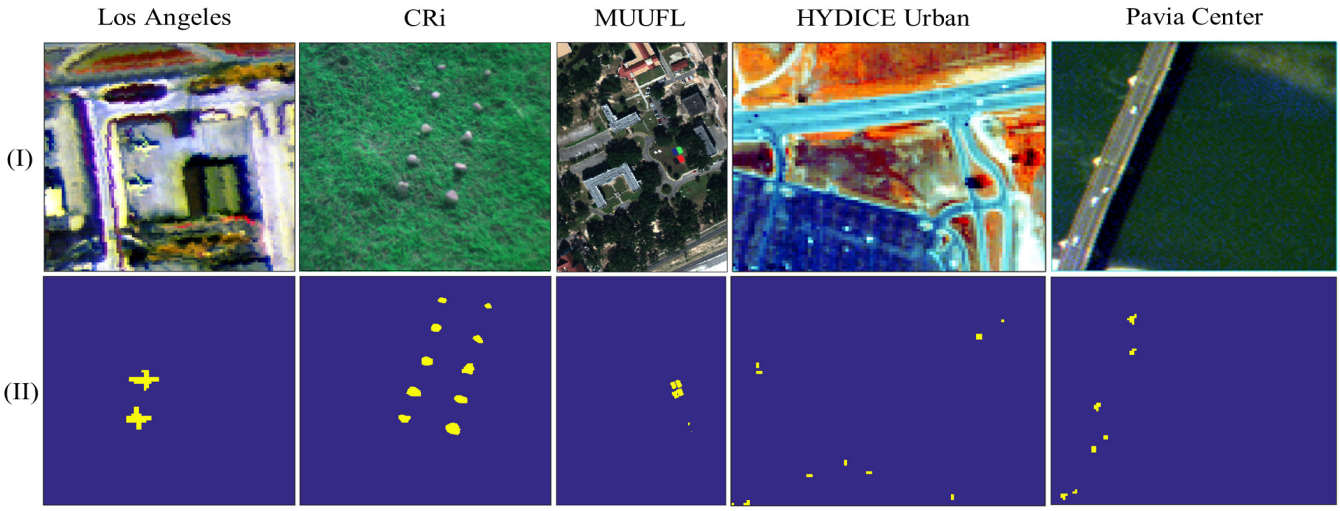


Fig. 4. Pseudocolor images and reference anomaly maps of five hyperspectral datasets. Row (I) gives the pseudocolor images. Row (II) gives the corresponding reference anomaly maps.

TABLE I
PARAMETERS' SETTING OF ALL METHODS FOR FIVE DATASETS

Methods	Parameters	Los Angeles	CRi	MUUFL	HYDICE Urabn	Pavia Center
RCRD	λ, τ, k	$10^{-2}, 10, 10$	$10^{-2}, 1, 1$	$10^{-5}, 10, 10$	$10^{-5}, 10^{2}, 10$	$1, 10^{-3}, 10$
RCRDW	λ, τ, k, γ	-, 10^{2}	-, 10^{2}	-, 10^{2}	-, 10^{2}	-,10^{2}
CRD	$window_size, \lambda$	$(13, 15), 1$	$(9, 15), 10^{-3}$	$(7, 15), 10^{-5}$	$(13, 15), 10^{-3}$	$(13, 15), 1$
GTVLRR	λ, β, γ	$0.5, 0.2, 005$				
SLDMM-AD	*patch size*, *sampling ratio*, z	$2 \times 2, 0.05, 10$				
KIFD	M, q, τ	$3\% \times N, 10^{3}, 300$				
2S-GLRT	$window_size$	$(13, 15)$	$(9, 15)$	$(13, 15)$	$(3, 15)$	$(3, 15)$
PTA	$\mu, rank, \alpha$ β, τ	$10^{-4}, 1, 10^{-4}$ $1, 1$	$10^{4}, 1, 10^{2}$ $10^{-4}, 1$	$10^{-2}, 1, 1$ $10^{-2}, 10^{-4}$	$10^{4}, 1, 10^{4}$ $10^{3}, 1$	$10^{-4}, 1, 1$ $10^{-1}, 10^{-1}$
RGAE	λ, S, n_hid	$10^{-2}, 10^{3}, 10^{2}$	$10^{-1}, 10^{3}, 10^{2}$	$10^{-4}, 150, 10^{2}$	$10^{-1}, 150, 10^{2}$	$10^{-2}, 150, 10^{2}$

C. Evaluation Indicators

The ROC curve is a well-known quantitative analysis method to evaluate the performance of detection algorithms, where the detection map gives continuous gray values [56]. It depicts the relationship between the false alarm rate P_{FA} and the detection rate P_D through a curve under a rectangular coordinate system, where P_{FA} is the $\boldsymbol{x}$-axis and P_D is the $\boldsymbol{y}$-axis. The area under the curve (AUC) gives the detection accuracy of methods (the larger the area, the higher the accuracy). P_{FA} and P_D are computed as

$$P_{\text{FA}} = \frac{N_{\text{BT}}}{N_B}, \quad P_D = \frac{N_{\text{TT}}}{N_T} \tag{31}$$

where N_{BT} is the number of background pixels marked as anomalous pixels, N_{TT} is the number of detected anomalous pixels, N_B is the total number of background pixels, and N_T is the total number of target pixels. If a method can quickly achieve the highest detection rate at a low false alarm rate (that is, the AUC is close to 1), this means that the method has excellent performance.

The box plot is a statistical way to evaluate the performance of anomaly detection methods. It draws the pixel values of both background and anomaly parts of the detection result map into a box, in a coordinate system. If the distance between the two boxes is large, the separation between background and anomalies is large. The shorter the background box, the better the background suppression degree. Good background suppression and high separation between background and anomalies reveal the excellent performance of an anomaly detection method.

D. Experimental Results

The anomaly detection maps of all the considered anomaly detection methods for the five datasets are shown in Fig. 5, and the ROC curves and box plots are displayed in

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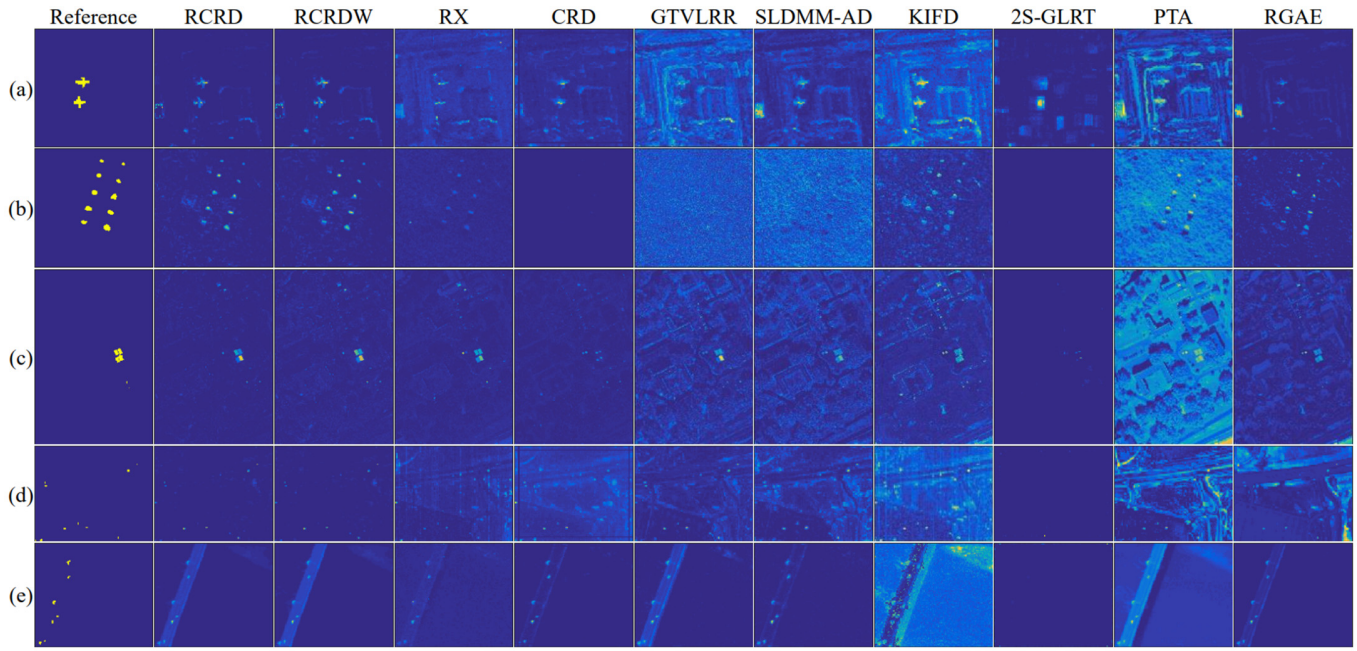


Fig. 5. Anomaly detection maps obtained by several anomaly detection methods for the five considered datasets: (a) Los Angeles, (b) CRi, (c) MUUFL, (d) HYDICE Urban, and (e) Pavia Center.

TABLE II
AUC AND RUNNING TIME OF SEVERAL ANOMALY DETECTION METHODS FOR THE FIVE CONSIDERED DATASETS

Dataset		RCRD	RCRDW	RX	CRD	GTVLRR	SLDMM-AD	KIFD	2S-GLRT	PTA	RGAE
Los Angeles	AUC (%)	<u>98.9628</u>	**99.0364**	84.0305	97.7905	88.7093	97.9536	97.8506	98.5100	93.7135	95.7885
	Time (s)	100.39	291.24	0.17	11.12	391.43	649.32	94.88	193.67	67.85	233.59
CRi	AUC (%)	**99.2227**	<u>99.2208</u>	91.3458	95.7112	63.3822	60.7895	97.0856	76.0632	89.3553	91.0933
	Time (s)	135.27	414.78	0.73	526.10	12910.66	2697.53	515.65	525.04	190.22	1651.31
MUUFL	AUC (%)	<u>99.9073</u>	**99.9109**	99.7668	95.3537	98.6781	99.1411	99.1622	76.9459	98.7150	97.9452
	Time (s)	224.04	682.41	0.55	57.48	1671.80	1468.00	242.35	987.34	96.31	1034.75
HYDICE Urban	AUC (%)	<u>99.9060</u>	**99.9164**	98.5668	94.9898	99.0362	97.7796	99.7156	86.7861	91.2165	81.2750
	Time (s)	136.42	261.58	0.35	17.62	314.77	463.23	68.76	42.35	49.98	150.11
Pavia Center	AUC (%)	<u>99.9734</u>	**99.9783**	99.8211	95.3709	99.6363	99.9243	84.6332	94.5988	99.9452	99.9451
	Time (s)	99.54	184.79	0.12	11.07	55.18	420.43	108.59	28.85	23.43	116.18

Figs. 6–10. In the box plots, the values distributed between $Q_1 + 1.5(Q_3 - Q_1)$ and $Q_3 + 1.5(Q_3 - Q_1)$ are shown, and the values beyond that range are ignored, where Q_3 and Q_1 are the upper quartile and the lower quartile of the box plot, respectively. The detection accuracy and the running time of all methods are given in Table II, where the best accuracy is in bold and the second one is underlined. The datasets adopted in experiments comprise three scenes with large targets and two scenes with small targets. Therefore, the experimental results are described from these two aspects.

1) Anomaly Detection Results for Large Targets: The first dataset (Los Angeles) is a small scene with two large targets, and the background is relatively complex. It can be seen from Fig. 5(a) that the targets in the anomaly maps of RCRD, RCRDW, CRD, SLDMM-AD, KIFD, and PTA are more obvious and complete than in the other methods, and the background is suppressed well by RCRD, RCRDW, 2S-GLRT, and RGAE. These also correspond to the results of the box plot in Fig. 6(b). Although the separation degree between the anomalies and background of KIFD is better than in the proposed methods, the background suppression degree of the proposed methods outperforms all the other methods, which indicates that the representation ability of the proposed models (RCRD and RCRDW) is strong. In addition, the ROC curves of RCRD and RCRDW are always higher than other methods when the false alarm rate ranges from 0 to 0.1, as shown in Fig. 6(a), and the AUCs of the proposed methods RCRD and RCRDW are 98.9628% and 99.0364%, respectively, which improves by 1.2% the result of CRD. Compared to RCRD, RCRDW can obtain more accurate results, but the running time is longer. This is because the RCRDW utilizes band features more flexibly by adding a feature weight, while RCRD does not require iterative computations.

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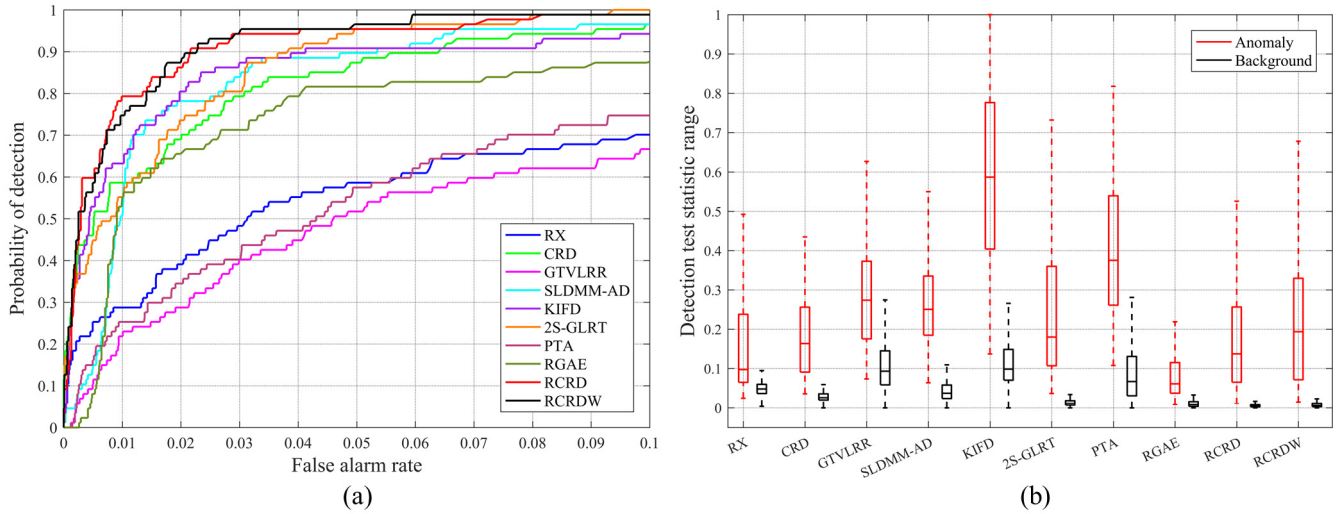


Fig. 6. ROC curve and box plots of several anomaly detection methods for the Los Angeles dataset. (a) ROC curve. (b) Box plots.

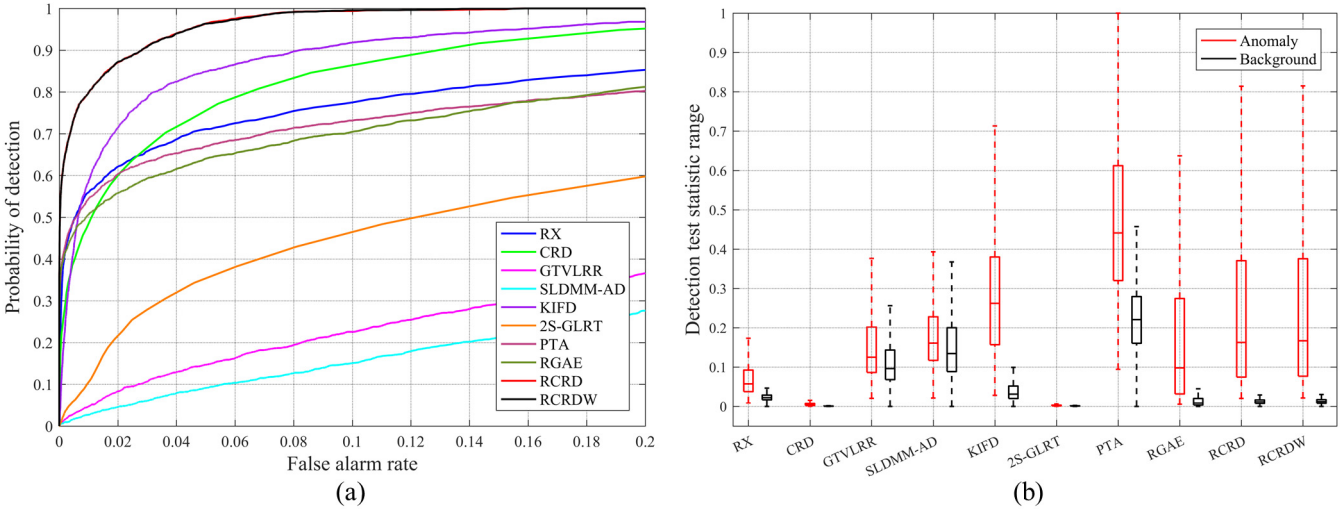


Fig. 7. ROC curve and box plots of several anomaly detection methods for the CRi dataset. (a) ROC curve. (b) Box plots.

The second dataset CRi is a large scene with two types of ground objects: grassland and stones. Intuitively, these anomalies may be easy to be detected. However, the targets almost cannot be seen in the anomaly maps of RX, CRD, GTVLRR, SLDMM-AD, and 2S-GLRT, as shown in Fig. 5(b), which means that these methods cannot distinguish the anomalies from the background. The targets are more obvious in the anomaly maps of RCRD, RCRDW, KIFD, PTA, and RGAE, but the proposed methods suppress the background more effectively. Furthermore, the same conclusion can be drawn from Fig. 7(b). RCRD and RCRDW highlight anomalies and suppress background simultaneously. The ROC curves reveal significantly different performances between the compared methods. Specifically, the ROC curves of RCRD and RCRDW reach the upper leftmost corner, followed by KIFD. When the false alarm rate is about 0.1, the probability of detection of the proposed methods yields 1, which is the highest one among all methods. The AUCs of the proposed methods (RCRD and RCRDW) are 99.2227% and 99.2208%, respectively, which improves by about 3.5% the results of CRD. The accuracy of RCRD is higher than that of RCRDW for the CRi dataset. A possible reason is that the RCRD model may be sufficient for this dataset, in which the ground objects are just grass and stones. Adding a feature weight may result in an unstable solution in this context.

The third dataset (MUUFL) is larger than the first one but smaller than the second one, with a relatively complex background. As shown in Fig. 5(c), the targets in the anomaly maps obtained by all methods are clear (except for CRD and 2S-GLRT), and the proposed methods detect the targets better than other methods. In addition, it can be seen from Fig. 8(b) that the separation degrees between anomalies and background for RX, GTVLRR, SLDMM-AD, KIFD, PTA, RGAE, RCRD, and RCRDW are almost the same, but the background suppression degree of the proposed methods and RX is better than that obtained by the other methods. The ROC curves reveal that the performance of RCRD, RCRDW, and RX is quite similar, and their ROC curves are always higher

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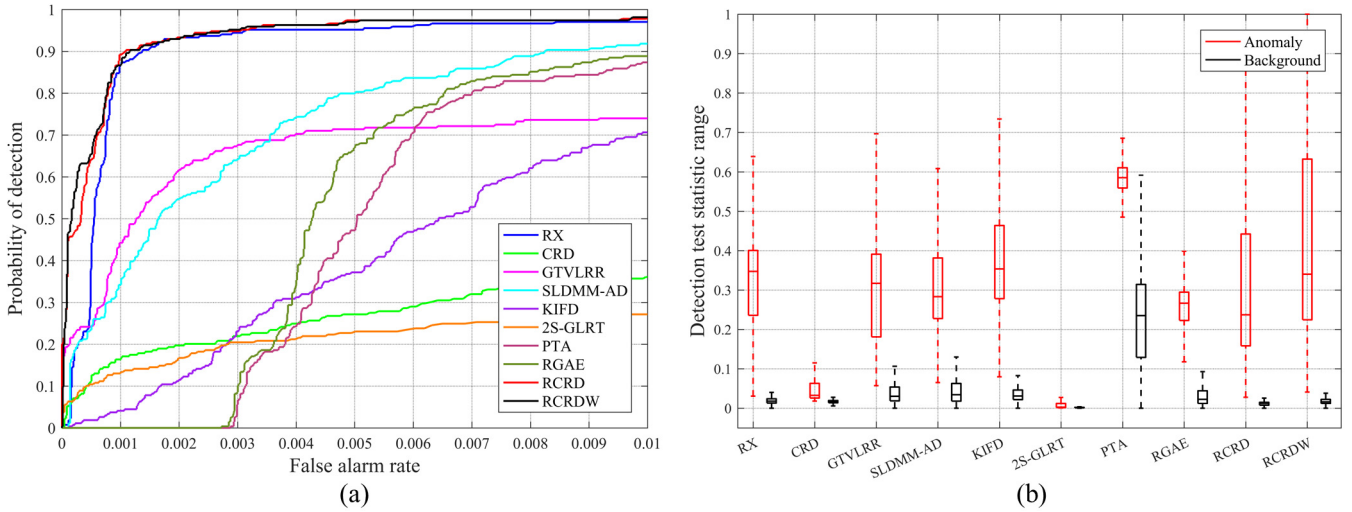


Fig. 8. ROC curve and box plots of several anomaly detection methods for the MUUFL dataset. (a) ROC curve. (b) Box plots.

than those obtained by other methods. In Table II, the AUCs of RX and the proposed methods (RCRD and RCRDW) are 99.7668%, 99.9073%, and 99.9109%, respectively. As a result, the accuracy of the proposed methods is better than that of RX and is increased by about 4.5% compared to CRD. The accuracy of RCRDW is slightly better than the one obtained by RCRD for this dataset.

From these experiments, it can be concluded that background complexity is not the main factor affecting detection accuracy. The CRi dataset is the simplest scene, but the accuracy of the algorithms with this scene is not the best. This is because the ground objects in the CRi image are natural, rather than man-made ones, which makes the difference between targets and background small, increasing the challenges for anomaly detection. As shown in the box-plot figures of the three datasets, the proposed methods suppress background and highlight anomalies effectively. This indicates that the proposed methods not only use spectral features flexibly to reconstruct the background by relaxing the constraints and adding a feature weight but also build a robust and adaptive dictionary for optimal representation of every pixel under test. In terms of detection accuracy and running time, except for the CRi dataset, RCRDW can obtain higher accuracy than RCRD for the other two datasets. The differences in accuracy between RCRD and RCRDW are small, while the running time of RCRD is much lower than that of RCRDW. In addition, the size of images has some influence on the running time of the proposed methods, but it is acceptable compared to other methods of the same type. Therefore, RCRDW is more suitable for applications requiring high accuracy, and RCRD is the best choice when the detection needs to be performed in a fast manner.

2) Anomaly Detection Results for Small Targets: Different from the Los Angeles, CRi, and MUUFL datasets, the HYDICE Urban and Pavia Center datasets are scenes in which targets only cover several pixels or even a pixel. Intuitively, these two datasets may be challenging for anomaly detection purposes. In fact, the targets in the HYDICE Urban image are almost invisible to the naked eye from Fig. 4(I). In the anomaly maps given in Fig. 5(d), it can be seen that the targets are visible except for 2S-GLRT. However, except for the proposed methods, it is difficult to distinguish targets from the background due to the relatively poor background suppression. As shown in Fig. 9, the proposed methods exhibit excellent background reconstruction performance. In terms of the ROC curve, the proposed methods are consistently better than other methods when the false alarm rate ranges from 0 to 0.02. The AUCs of RCRD and RCRDW are 99.9060% and 99.9164%, respectively, which is about 5% better compared to CRD. RCRDW obtains more accurate results than RCRD, but RCRD is faster.

The targets of the Pavia Center dataset are bigger than those of the HYDICE Urban dataset, and the background is more simple. In Fig. 5(e), it can be seen that the performance of all methods is good for this scene (except for KIFD and 2S-GLRT). In Fig. 10(b), it can be seen that the background suppression effects of all methods are similar, but the separation degree between anomalies and background of GTVLRR, SLDMM-AD, PTA, RGAE, RCRD, and RCRDW is better than those achieved by other methods. In addition, Fig. 10(a) reveals that the ROC curve of the proposed methods is always higher than that of the other methods and is the first one to reach an optimal probability of detection when the false alarm rate ranges from 0 to 0.01. The AUCs of RCRD and RCRDW are 99.9734% and 99.9783%, respectively, which represents an increase of about 4.5% compared to CRD. The accuracy of RCRD and RCRDW is quite similar, while the running time of RCRD is about half that of RCRDW.

The experiments of these two scenes containing small man-made targets reveal that background complexity has some influence on detection accuracy, i.e., the simpler the background, the higher the accuracy. Similar to the conclusions drawn from our experiments with scenes containing larger targets, the representation ability of the proposed methods, RCRD and RCRDW, is quite strong, and the background is greatly suppressed. Moreover, RCRDW exhibits higher detection accuracy (with slightly higher running time) than RCRD.

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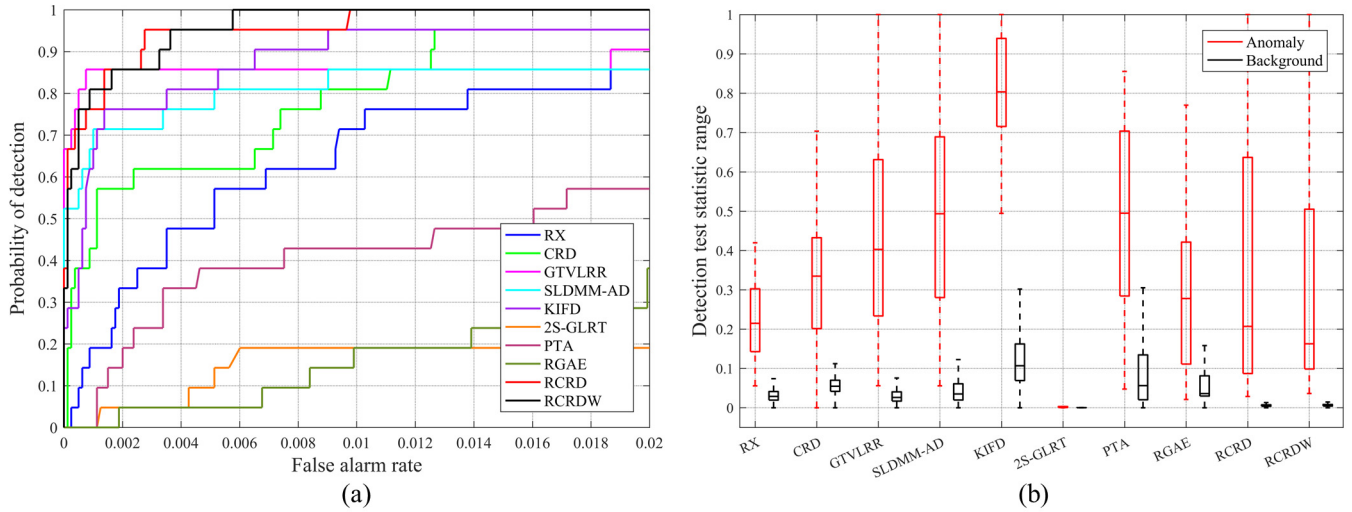


Fig. 9. ROC curve and box plots of several anomaly detection methods for the HYDICE Urban dataset. (a) ROC curve. (b) Box plots.

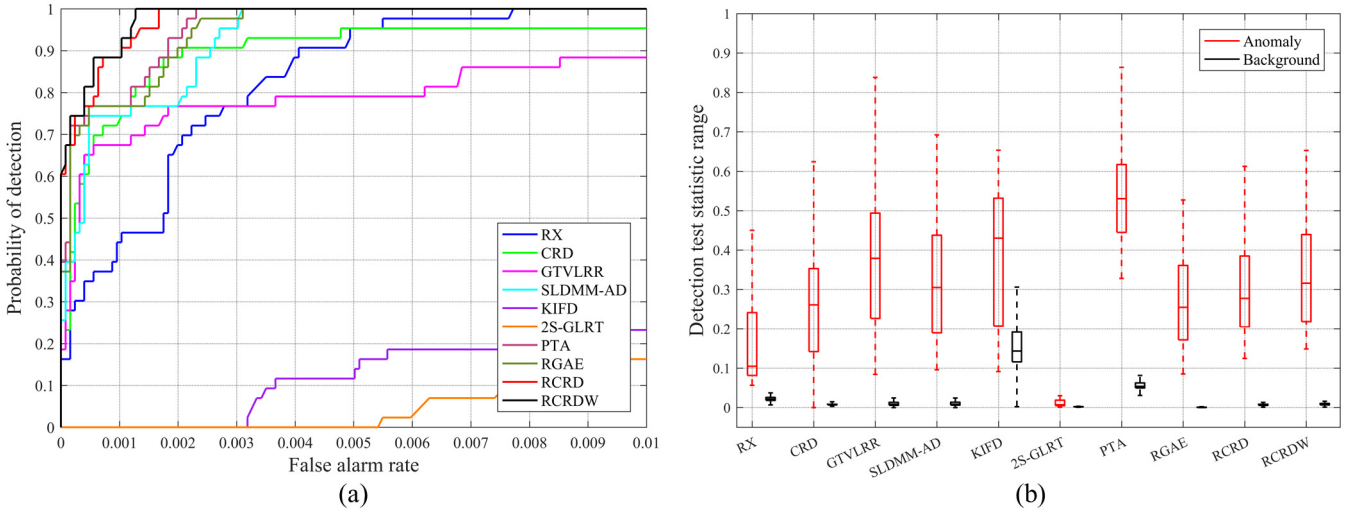


Fig. 10. ROC curve and box plots of several anomaly detection methods for the Pavia Center dataset. (a) ROC curve. b) Box plots.

In summary, although the detection of small targets is intuitively more challenging, the experimental results obtained for different datasets show that most algorithms exhibit better performance when detecting small targets. There are two possible reasons for this peculiar behavior. One is that the edges of large targets may contain mixing effects (between background and target) that complicate the detection task. Another reason is that many anomaly detection methods are designed based on the characteristics of small targets, so the generalization performance of methods for large targets is poor. In comparison, the proposed methods can be applied to detect both types of targets with excellent performance in terms of background suppression. In addition, as can be seen in the box plots, the separation between background and anomaly of the proposed method is not as effective as the background suppression. This is because strong representation capability makes the reconstruction residuals of the anomalous part also relatively small. Furthermore, RCRDW generally performs better than RCRD (except for extremely simple datasets), which also indicates the importance of incorporating the differences in features for anomaly detection purposes. RCRD runs faster (about half the time of RCRDW), which makes it more suitable for fast detection missions.

IV. Discussion

In this section, three different aspects of the proposed methods are discussed to analyze their performance more comprehensively, i.e., the influence of band selection on the proposed methods, the effectiveness of nonglobal dictionaries, and sensitivity analysis of the proposed methods to input parameters.

A. Influence of Band Selection on RCRD and RCRDW

Both RCRD and RCRDW are computed in a band-by-band fashion, which is time-consuming for anomaly detection. Therefore, this section explores the influence of band selection on the proposed algorithms. The proposed methods, RCRD

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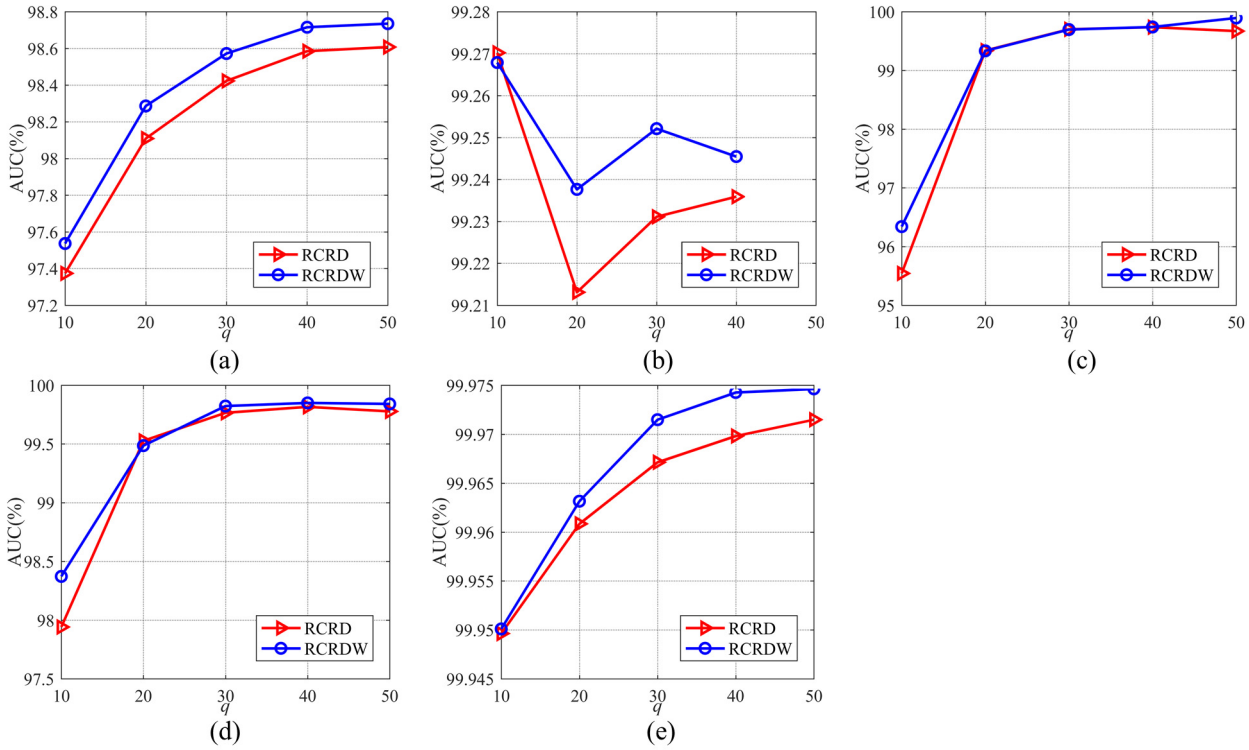


Fig. 11. Changes in the accuracy of RCRD and RCRDW when using different numbers of selected bands for five datasets: (a) Los Angeles, (b) CRi, (c) MUUFL, (d) HYDICE Urban, and (e) Pavia Center.

TABLE III
AUC AND RUNNING TIME OF RCRD-BS, RCRDW-BS, RCRD, AND RCRDW FOR THE FIVE CONSIDERED DATASETS

Dataset		RCRD-BS	RCRDW-BS	RCRD	RCRDW
Los Angeles	q	50	50	205	205
	AUC(%)	98.6074	98.7349	98.9628	**99.0364**
	Times(s)	**26.39**	79.37	100.39	291.24
CRi	q	10	10	46	46
	AUC(%)	**99.2702**	99.2679	99.2227	99.2208
	Times(s)	**35.35**	106.03	135.27	414.78
MUUFL	q	40	50	64	64
	AUC(%)	99.7366	99.8929	99.9073	**99.9109**
	Times(s)	**145.25**	587.34	224.04	682.41
HYDICE Urban	q	40	40	175	175
	AUC(%)	99.8162	99.8496	99.9060	**99.9165**
	Times(s)	**22.97**	63.55	136.42	261.58
Pavia Center	q	50	50	102	102
	AUC(%)	99.9715	99.9746	99.9734	**99.9783**
	Times(s)	**31.77**	96.49	99.54	184.79

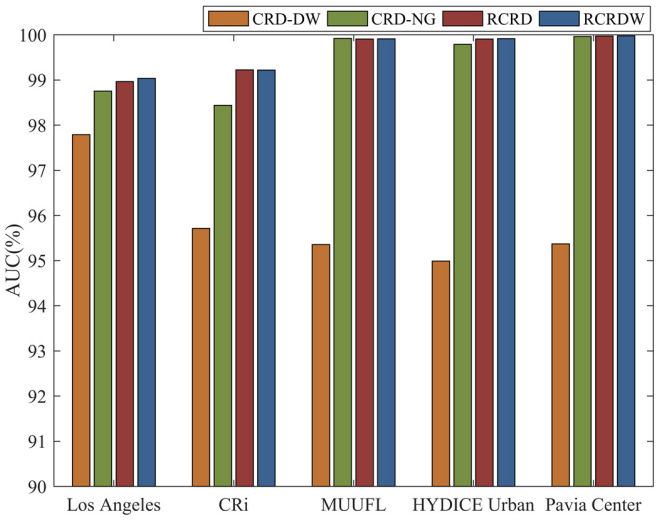


Fig. 12. Accuracy of CRD-DW, CRD-NG, RCRD, and RCRDW for the five considered datasets. CRD-DW is the combination of CRD and dual window, and CRD-NG is the combination of CRD and the nonglobal dictionary method proposed in this article.

and RCRDW, combined with the band selection method in Section II-D are renamed RCRD-BS and RCRDW-BS. If RCRD-BS and RCRDW-BS can detect anomalies well with a few bands, the band selection version of the proposed methods will be more practical. The setting range of the number of bands q for RCRD and RCRDW is $\{10, 20, 30, 40, 50\}$ for all datasets except for the CRi dataset, which only has 46 bands. Band selection is carried out by using the method described in Section II-D. The experimental results obtained for the five considered datasets are displayed in Fig. 11. It can be seen that the accuracy of RCRD and RCRDW increases for all datasets with the increase in the number of bands, except for the CRi dataset, and the growth rate is slower and slower. When the number of bands is about 50, the accuracy of RCRD and RCRDW for the Los Angeles, MUUFL, HYDICE Urban, and Pavia Center is the highest. The performance of RCRD

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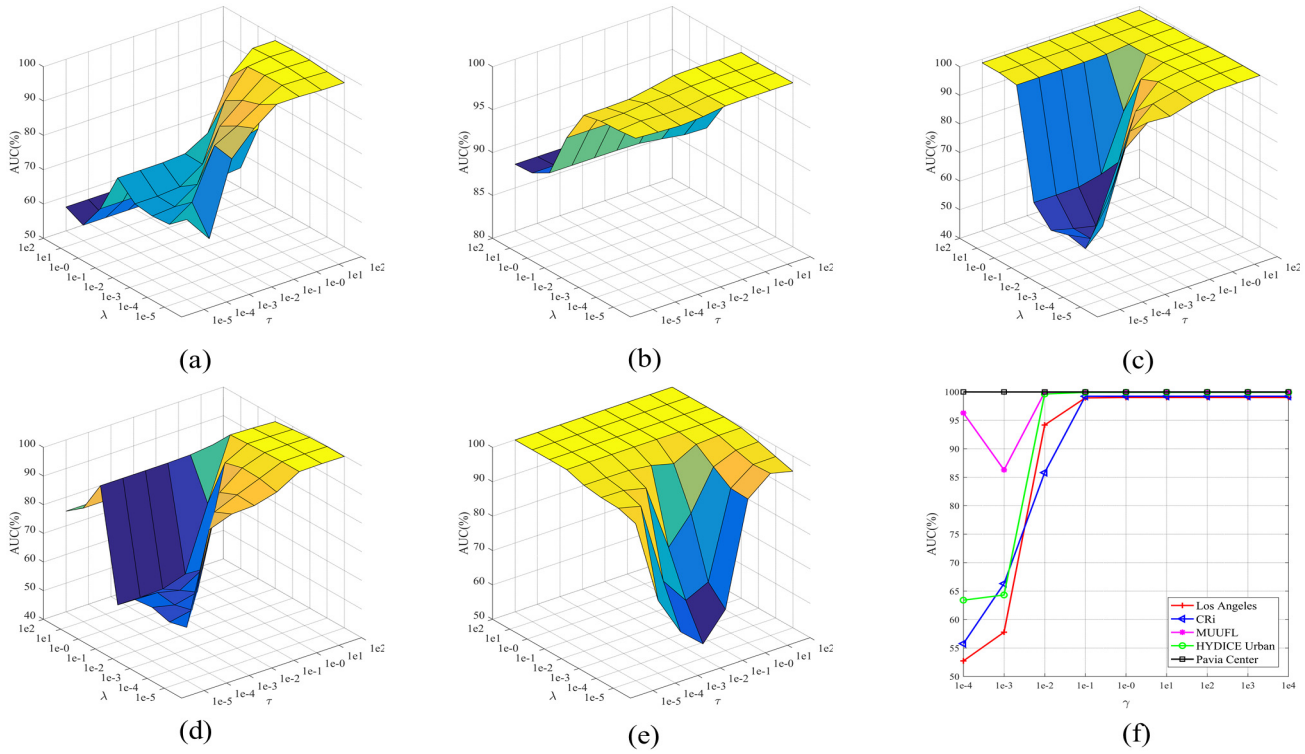


Fig. 13. Sensitivity analysis of RCRD and RCRDW to λ, τ, and γ. (a) Los Angeles, (b) CRi, (c) MUUFL, (d) HYDICE Urban, and (e) Pavia Center show the sensitivity analysis of RCRD to λ and τ, which is similar to that of RCRDW. (f) Sensitivity analysis of RCRDW to γ.

and RCRDW for the CRi dataset is different, with the accuracy of RCRD and RCRDW first decreasing and then increasing slowly with the increase in the number of bands. When the number of bands is 10, the accuracy of RCRD and RCRDW is the highest. In addition, it should be noticed that the accuracy of RCRDW is always higher than that of RCRD for all datasets when the number of bands is increased. This indicates that the adaptive weight of features can further improve the performance of the model.

To compare the performance of RCRD-BS, RCRDW-BS, RCRD, and RCRDW, the optimal results of these four methods are listed in Table III. As shown in Table III, band selection versions of the proposed methods significantly reduce the running time at the (slight) expense of accuracy. RCRD-BS and RCRDW-BS even outperform RCRD and RCRDW without band selection with only ten bands for the CRi dataset. A possible reason is that the information comprised of ten bands is enough for this dataset, where the ground objects are just grass and stones. More bands will only result in information redundancy and weaken the performance of the algorithms. Therefore, the proposed methods (combined with band selection) can not only reduce the running time but also achieve better detection results for simple datasets. Even for complex datasets, the algorithm performs quite well when 40 bands are selected.

B. Effectiveness of the Nonglobal Dictionary

To support the detection model of the proposed RCRD and RCRDW methods, a nonglobal dictionary construction method is designed by using MSC and KNN methods. The nonglobal dictionary is an extension of a global dictionary in pixel-by-pixel anomaly detection methods. The global dictionary is proposed in [28], which is proven to be efficient for global anomaly detection models. The proposed nonglobal dictionary method selects k atoms in the global dictionary, which are similar to the pixel under test to participate in the reconstruction. To test the effectiveness of the nonglobal dictionary, CRD with dual window (CRD-DW), CRD with a nonglobal dictionary (CRD-NG), RCRD, and RCRDW are used to detect anomalies in the five considered datasets, and the results are reported in Fig. 12. It can be seen that the detection performance of CRD-NG for the five datasets has been significantly improved compared to that of CRD-DW. This shows that the proposed nonglobal dictionary is more reliable and stable, which also requires almost no parameter tuning. Moreover, the detection performance of RCRD and RCDW on the five datasets is always better than that of CRD-NG, especially in the Los Angeles and CRi scenes, which reveals the advantages of the detection model in the proposed RCRD and RCRDW methods, i.e., features should keep both difference and similarity.

C. Sensitivity Analysis to Input Parameters

The proposed methods have four parameters, λ, τ, and k for both RCRD and RCRDW, and γ for RCRDW. In this section, a sensitivity analysis of RCRD and RCRDW to input parameters is conducted.

1) Setting of λ, τ, and γ: Our experiments show that the sensitivity analysis of RCRD and RCRDW to λ and

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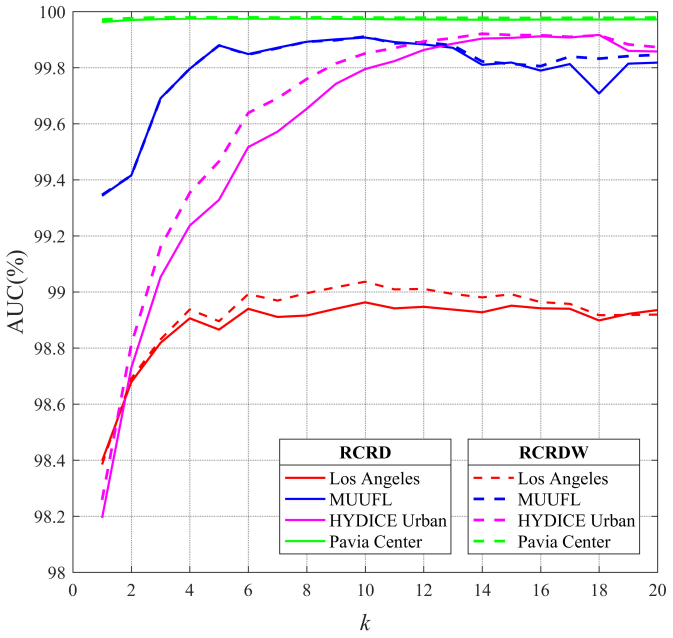


Fig. 14. Sensitivity analysis of RCRD and RCRDW to the number of nonglobal dictionary atoms k.

τ is similar, so Fig. 13(a)–(e) displays the change in detection accuracy of RCRD with different values of λ and τ. It can be seen that RCRD is not sensitive to λ, but it is sensitive to τ for the CRi dataset. When λ is set in the range {1e-3, 1e-4}, and τ is set in the range {1e1, 1e2}, the accuracy of RCRD and RCRDW of the Los Angeles, CRi, MUUFL, and HYDICE Urban datasets is high. For the Pavia Center dataset, the AUC is high when λ is set in {1e0, 1e1}. Therefore, the recommend values for λ and τ are {1e-3, le0} and 1e2, respectively. As shown in Fig. 13(f), when γ is larger than 1e-1, the performance of RCRDW is good and stable. In practical applications, γ is recommended to be set to 1e2 for RCRDW.

2) Setting of k: In the previous experiments, k is set to 10 for every dataset, except for CRi, in which the global dictionary only has one atom. This part analyzes the influence of different values of k on the performance of RCRD and RCRDW. Fig. 14 shows that the accuracy of RCRD and RCRDW first goes up and then keeps the balance for Los Angeles, MUUFL, and HYDICE Urban datasets with the increase in k. For the Pavia Center dataset, the accuracy of RCRD and RCRDW is not sensitive to k. Therefore, without any prior information, k is recommended to be set to 10 for most datasets, and all atoms of the global dictionary will participate in the representation when the number of dictionary atoms is less than 10. In addition, the accuracy of RCRDW is always higher than that of RCRD for all datasets with the increase in k, which also proves the superiority of our RCRDW model.

V. Conclusion

In this article, a new anomaly detector based on RCRD is proposed. The new RCRD can make full use of each spectral information to detect anomalies under a nonglobal background dictionary. The proposed RCRD conducts the CR on each spectral feature to use both differences and similarities among features for accurate detection. Moreover, a variation of the RCRD method constrained by a weight (RCRDW) further improves the representation ability of the model. The new dictionary, which is built by using the KNN on a global dictionary, is proven to be more efficient than the widely used dual-window dictionary. In addition, the influence of band selection on the newly proposed methods has been also comprehensively studied. Experimental results on five real datasets reveal that the proposed methods outperform other state-of-the-art anomaly detection models, achieving detection accuracies over 99% for scenes with both small and large targets. Our newly proposed RCRD and RCRDW methods, when combined with band selection, also achieve outstanding results with less running time. In the future, we will conduct efficient implementations of both RCRD and RCRDW using specialized hardware, such as graphics processing units (GPUs).

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